



An Internship Report
Of
Data Analyst

Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P

In partial fulfilment of the requirement of degree of
B.Tech (CSE)

BY

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Batch :- 2026-27

CANDIDATE'S DECLARATION

I, ADAMYA DIXIT, do hereby declare that the internship report titled "Internship Report on Data Analytics" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science Engineering. All the references have been duly acknowledged, and I have not incorporated any material from any other source without citation. I affirm that this report is entirely my own work and has not been submitted, in full or in part, to any other institute for the award of any degree or diploma, and I shall be solely responsible for any copyright violation arising in this regard.

Signature

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to BeeSkilled for giving me the opportunity to undertake this six-week internship in the Data Analyst domain. This internship exposed me to practical, real-world data analysis practices and allowed me to apply the concepts learned during my academic curriculum in a hands-on setting.

I am thankful to IILM University, Greater Noida, U.P. and the School of Computer Science and Engineering for providing the academic foundation and encouragement that made this internship possible. I extend my heartfelt thanks to my faculty members and academic support staff for their guidance, constructive feedback, and continued support throughout this internship period.

I am also grateful to everyone at BeeSkilled who supported me during this internship for sharing their knowledge and insights on various aspects of Data Analytics. Finally, I express my deepest gratitude to my family for their constant encouragement and support throughout this journey.

Signature

Student Name: ADAMYA DIXIT
Roll No.: 25SCS1003003698

The certificate below was issued by BeeSkilled on 28 August 2026, confirming successful completion of the 6-week online internship program in the Data Analyst domain.



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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Data Analytics is the process of examining raw data with the objective of drawing conclusions, identifying patterns, and supporting informed decision-making. In today's increasingly data-driven business environment, organizations depend on the structured analysis of their operational data to understand performance, identify inefficiencies, and plan future strategy. Data Analytics as a discipline draws upon a range of tools and techniques, from spreadsheet-based analysis to programming-based data manipulation and visual, dashboard-driven reporting.

Before any meaningful analysis can be carried out, raw data must first be organized and cleaned. Real-world datasets frequently contain missing values, inconsistent formatting, and duplicate records, so data cleaning forms a foundational step that determines the reliability of every subsequent stage of analysis. Once cleaned, data can be explored using tools such as Microsoft Excel for spreadsheet-based analysis, Python together with the Pandas library for programmatic data manipulation, and business intelligence platforms such as Power BI for building dashboards and calculated business measures that communicate findings to stakeholders.

This report documents a six-week internship in the Data Analyst domain undertaken with BeeSkilled, during which these concepts were studied and applied through four weekly modules: Excel-based sales analysis, Python/Pandas-based dataset exploration, Python-based data wrangling and cleaning, and a concluding data-visualization and business-analytics capstone project. The internship provided an opportunity to connect the theoretical foundation gained through the B.Tech CSE curriculum with practical, hands-on data analysis work, and to understand how raw business data is progressively transformed into decision-ready insight.

4.2 Organization Profile

BeeSkilled is an online internship and skill-development platform operating under the tagline "Tech and Internships," offering structured, project-based internship programs designed to build participants' practical knowledge and hands-on industry experience. The organization is registered with AICTE (All India Council for Technical Education) and

MSME (Ministry of Micro, Small and Medium Enterprises), as indicated on its official internship communication.

The internship documented in this report was undertaken in the Data Analyst domain and structured as a six-week online program combining weekly curriculum modules with hands-on assignments and a concluding capstone project. The program was headed by Ms. Simranjeet Kaur, Founder & Program Head, and the completion certificate was issued upon successful completion of the assigned coursework and capstone project.

4.3 Problem Statement

Organizations generate large volumes of raw data through their everyday operations, but this data carries little practical value until it is processed into a form that supports decision-making. The core problem addressed throughout this internship was learning how to convert raw, unstructured, or messy datasets into clear, actionable business insight using a structured analytical workflow:

**Raw Data → Data Cleaning → Data Transformation → DataAnalysis → Visualization →
Insights → Recommendations**

This workflow was applied practically across the internship: in Excel, by summarizing a large orders dataset into pivot-table-based business metrics; in Python and Pandas, by exploring a transactional dataset and separately cleaning a messy exercise-tracking dataset with missing and duplicate records; and in the Week 4 capstone project, by carrying a raw sales-transaction dataset through cleaning, validation, and exploratory business analysis to arrive at concrete, stakeholder-ready recommendations.

4.4 Project Objectives

- Understand the fundamentals of Data Analytics and its role in business decision-making.
- Learn Excel-based data analysis, including pivot tables and aggregation of large transactional datasets.
- Analyze revenue, sales, and category-wise business data using pivot-table dashboards.
- Learn Python fundamentals and use the Pandas library for structured dataset exploration.

- Apply core Pandas methods — `head()`, `info()`, `shape`, `describe()` — to inspect and summarize a dataset.
- Clean real-world datasets by identifying and handling missing values and duplicate records.
- Convert inconsistent date fields into proper datetime formats and impute missing values appropriately.
- Create derived/calculated columns from existing data for further analysis.
- Filter datasets based on business-relevant conditions.
- Understand dashboard development and DAX-based calculated measures using Power BI.
- Learn the principles of data storytelling and structuring insights for a stakeholder audience.
- Develop an end-to-end data analytics workflow through a capstone project, from raw data to business recommendations.

4.5 Scope of the Project

The scope of this internship was limited to structured, dataset-based analysis using sample and practice datasets provided as part of the BeeSkilled curriculum; it did not involve analysis of live production or proprietary company data. Within this scope, the internship covered:

- Handling and organizing large structured business datasets (sales and order data).
- Pivot-table-based analysis and business-metric summarization using Microsoft Excel
- Dataset inspection and exploration using Python and the Pandas library.
- Data cleaning and transformation, including missing-value handling, duplicate removal, and date-format correction.
- Creation of derived columns and condition-based filtering of records.
- Data validation and consistency checks across related numeric fields.

- Business intelligence measure-building (DAX) and dashboard concepts using Power BI.
- Data storytelling — translating analytical findings into recommendations for stakeholders.

4.6 Technologies and Tools Used

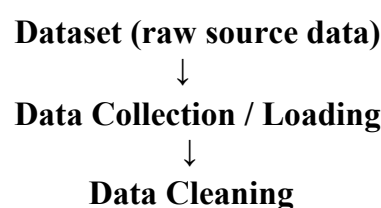
Table 4.1 summarizes the primary tools and technologies used during the internship, along with their purpose and how each was applied.

Tool	Purpose	Application during Internship
Microsoft Excel	Spreadsheet-based data organization and analysis	Built a pivot-table dashboard analyzing total revenue, sales by category, yearly trends, and segment-wise revenue from a 51,290-record orders dataset
Python	General-purpose programming for data analysis tasks	Used for dataset loading, inspection, and data-cleaning scripts across Weeks 2 and 3
Pandas	Data manipulation — reading, inspecting, cleaning, and transforming data	Inspected a 200-row sales dataset using <code>head()/info()/describe()</code> ; cleaned a 32-row exercise dataset by handling missing values, duplicates, and date formatting; created a derived column and filtered records
Power BI	Business intelligence dashboarding and calculated (DAX) measures	Built DAX measures — Total Sales, Total Profit, Total Units Sold, Profit Margin, Average Order Value, and Loss-Making Orders — for the Week 4 capstone dashboard

Table 4.1: Technologies and Tools Used

4.7 Data Analytics Workflow

Rather than a software-engineering system architecture, the internship followed a Data Analytics workflow that moves from raw data collection through cleaning, transformation, and analysis, to visualization and, finally, actionable recommendations. This workflow, shown in Figure 4.1, was applied consistently across all four weeks of the internship.



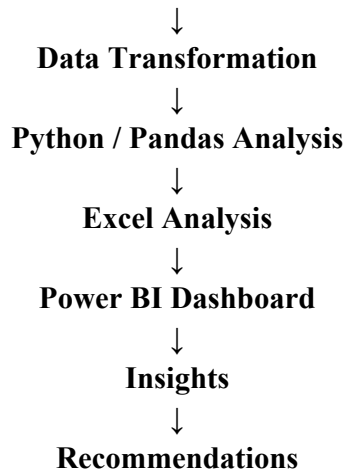


Figure 4.1: Data Analytics Workflow

4.8 Methodology

This section documents the methodology followed during each week of the internship, along with the specific work performed and the results obtained.

4.8.1 Week 1: Excel Data Analysis

Week 1 of the internship focused on Excel fundamentals for data analysis, covering spreadsheet navigation, aggregation formulas, pivot tables, and data summarization techniques.

The Week 1 assignment involved building a pivot-table-based analysis of sales performance from the "Global Superstore" Orders dataset, which contained approximately 51,290 records across 24 columns. Category-wise sales, shown in Figure 4.2, were led by Technology at approximately \$4.74M, followed by Furniture at approximately \$4.11M and Office Supplies at approximately \$3.79M. Yearly sales showed a consistent upward trend, rising from approximately \$2.26M in 2012 to approximately \$4.30M in 2015.

<i>Sales by Category</i>	<i>Total Sales</i>
Technology	\$4.74M
Furniture	\$4.11M
Office Supplies	\$3.79M

<i>Yearly Sales Trend</i>	<i>Total Sales</i>
2012	\$2.26M

<i>Yearly Sales Trend</i>	<i>Total Sales</i>
2013	\$2.68M
2014	\$3.41M
2015	\$4.30M

Figure 4.2: Excel Pivot Table — Sales Performance Analysis (Category-wise and Yearly)

Customer-segment-wise revenue was also analyzed using pivot tables, shown in Figure 4.3. The Consumer segment generated the highest revenue at approximately \$6.51M, followed by Corporate at approximately \$3.82M and Home Office at approximately \$2.31M.

<i>Customer Segment</i>	<i>Total Sales</i>
Consumer	\$6.51M
Corporate	\$3.82M
Home Office	\$2.31M

Figure 4.3: Segment-wise Revenue Analysis

This exercise demonstrated how Excel pivot tables can be used to organize a large raw transactional dataset into summarized, decision-ready business metrics using SUM-based aggregation, without writing any code.

4.8.2 Week 2: Python/Pandas Data Analysis

Week 2 of the internship curriculum covered Python and the Pandas library for structured dataset analysis. Topics covered included loading a dataset with Pandas, inspecting its structure and data types, and generating descriptive statistics. The assigned exercise for this week was to load a sales dataset and perform an initial exploratory inspection of it.

A 200-row, 10-column sales dataset was loaded using Pandas, with columns `order_id`, `customer_name`, `order_date`, `category`, `sub_category`, `product_name`, `quantity`, `unit_price`, `total_price`, and `region`. The dataset was inspected using `df.shape` and `df.info()`, confirming its dimensions and column data types, as shown in Figure 4.4.

```
>>> df.shape
(200, 10)

>>> df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
```

```

Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype  -
---  ---
0   order_id               200    non-null  int64
1   customer_name          200    non-null  object
2   order_date             200    non-null  object
3   category               200    non-null  object
4   sub_category           200    non-null  object
5   product_name           200    non-null  object
6   quantity               200    non-null  int64
7   unit_price             200    non-null  float64
8   total_price            200    non-null  float64
9   region                 200    non-null  object

```

Figure 4.4: Dataset Structure Summary (df.shape and df.info() Output)

Figure 4.5 shows the first five records of the dataset retrieved using the `df.head()` method, giving an initial view of the raw transactional data prior to analysis, along with the list of column names retrieved using `df.columns`.

```

>>> df.columns.tolist()
['order_id', 'customer_name', 'order_date', 'category',
 'sub_category', 'product_name', 'quantity', 'unit_price',
 'total_price', 'region']

>>> df.head()
   order_id customer_name order_date    category  quantity
0      1000    <customer>    <date>    <category>         ..
1      1001    <customer>    <date>    <category>         ..
2      1002    <customer>    <date>    <category>         ..
3      1003    <customer>    <date>    <category>         ..
4      1004    <customer>    <date>    <category>         ..

```

Figure 4.5: Sample Dataset Records (df.head() and df.columns Output)

Figure 4.6 shows descriptive statistics generated using the `df.describe()` method for the numeric columns — `quantity`, `unit_price`, and `total_price` — giving a quick statistical overview of the dataset's central tendency and spread. The mean quantity per order was 4.91, the mean unit price was 2421.495, and the mean total price was 12,101.535, with a maximum total price of 47,940 recorded in the dataset.

```

>>> df.describe()
      quantity  unit_price  total_price
count  200.000000    200.000000    200.000000
mean     4.910000    2421.495000    12101.535000
std       ...           ...           ...
min       ...           ...           ...
max       ...           ...           47940.000000

```

Figure 4.6: Statistical Summary of the Dataset (df.describe() Output)

This exercise demonstrated how the combination of `shape`, `info()`, `head()`, and `describe()` provides a fast, reliable first impression of an unfamiliar dataset — confirming its size, column types, sample content, and numeric distribution before any deeper analysis is attempted.

4.8.3 Week 3: Python and Data Wrangling

Week 3 focused on data wrangling using Python and Pandas — identifying and resolving data-quality issues such as missing values, duplicate records, and inconsistent date formatting, and creating new calculated columns. The main assignment for the week was to clean a messy dataset in Pandas and export the cleaned result.

A workout/activity dataset of 32 rows and 5 columns (Duration, Date, Pulse, Maxpulse, Calories) was loaded and inspected. The inspection revealed one duplicate row, one missing Date value, and two missing Calories values, as summarized in Figure 4.7.

```
Original dataset      : 32 rows x 5 columns
Columns              : Duration, Date, Pulse, Maxpulse, Calories

Data quality issues found:
- Missing Date values      : 1
- Missing Calories values : 2
- Duplicate rows           : 1

Cleaning actions performed:
- Removed 1 duplicate row
- Converted Date column to proper datetime format
- Filled missing Date value using interpolation
- Filled missing Calories values using the column mean
- Added derived column: Calories_Per_Minute
  Calories_Per_Minute = Calories / Duration

Cleaned dataset      : 31 rows x 6 columns
```

Figure 4.7: Data Cleaning / Transformation Summary

Following the cleaning steps, the Date column — originally stored as inconsistent text — was converted into a proper datetime format, and the single missing Date entry was resolved using interpolation. The two missing Calories values were filled using the column mean, after which the dataset contained no remaining missing values. A new calculated column, `Calories_Per_Minute`, was created by dividing Calories by Duration for each record, and the resulting dataset (31 rows, 6 columns after the duplicate was dropped and the new column added) was exported as `cleaned_data.csv`.

As a further exercise, the cleaned dataset was filtered to retain only records where Calories exceeded 300, and the resulting subset was exported separately as `filtered_data.csv`. This week's work demonstrated the practical importance of data cleaning before analysis: inconsistent formatting and missing values can silently distort downstream calculations if

not corrected first, and verifying data quality is a necessary first step before any further analysis or filtering.

4.8.4 Week 4: Data Visualization and Capstone Project

Week 4 covered Power BI dashboarding fundamentals — calculated (DAX) measures and business-metric visualization — together with the principles of data storytelling: structuring analytical findings into insights and recommendations for a stakeholder audience. The week concluded with an End-to-End Business Analytics capstone project applying the full data-analytics workflow to a financial/sales transaction dataset.

The capstone dataset contained 700 records across 16 original columns, including Segment, Country, Product, Discount Band, Units Sold, Gross Sales, Discounts, Sales, COGS, Profit, Date, Month, and Year.

Data cleaning and validation: inspection of the dataset revealed 53 records with a blank Discount Band. Investigation of these records showed that they carried zero discount values, and they were accordingly classified as "No Discount" rather than being dropped, preserving valid transaction data. No duplicate records were found. The dataset's internal numeric consistency was also verified across all 700 rows — confirming that $\text{Units Sold} \times \text{Sale Price}$ equalled Gross Sales, $\text{Gross Sales} - \text{Discounts}$ equalled Sales, and $\text{Sales} - \text{COGS}$ equalled Profit — and no data-integrity issues were found. A Year-Month helper field was created to support the monthly trend analysis, and the cleaned dataset was exported as `Week_4_Cleaned_Data.csv`.

Following cleaning and validation, exploratory business analysis was carried out and a Power BI dashboard was built using calculated DAX measures, summarized in Figure 4.8.

<i>Business KPI</i>	<i>Value</i>
Total Sales	\$118,726,350
Total Profit	\$16,893,702
Total Units Sold	1,125,806
Average Order Value	\$169,609
Profit Margin	14.2%
Total Orders	700

Figure 4.8: Power BI Business Performance Dashboard — Key KPI Measures

DAX measures created for the dashboard included Total Sales, Total Profit, Total Units Sold, Profit Margin, Average Order Value, and Loss-Making Orders. Key findings from the exploratory and dashboard-driven analysis were as follows:

- Government was the strongest segment by profit, contributing approximately \$11.39M. Enterprise was the only segment to record an overall loss, at approximately –\$614,546.
- The USA recorded the highest sales among all countries, at approximately \$25.03M.
- Paseo was the strongest-performing product in both sales and profit, while Carretera was the weakest in both.
- 58 of the 700 orders (approximately 8.3%) were loss-making, and these were predominantly associated with Medium or High discount bands.
- October and December 2014 were among the strongest months by sales, while September 2013 was the weakest month in the analyzed period.

Based on these findings, the following recommendations were prepared for a stakeholder audience:

- Review discount-approval policies, given the association between higher discount bands and loss-making orders.
- Introduce minimum-margin controls for heavily discounted deals.
- Investigate the causes of Enterprise segment's overall loss before scaling the segment further.
- Investigate Carretera's pricing and cost structure, given its weak performance on both sales and profit.
- Plan resources and inventory around the observed seasonal (October–December) demand peak.
- Investigate the unusually weak September 2013 performance to identify any underlying cause.

This capstone project brought together every skill developed over the internship — data cleaning and validation, exploratory analysis, and BI-driven measure-building — into a single end-to-end workflow, from a raw transactional dataset to a set of stakeholder-ready business recommendations.

4.9 Expected Outcomes / Learning Outcomes

- Practical experience with Excel-based data analysis, including pivot tables and business-metric summarization from large datasets.
- Working knowledge of Python fundamentals and the Pandas library for dataset inspection and exploration.
- Hands-on experience with data cleaning: identifying and handling missing values, duplicate records, and inconsistent date formats.
- Ability to create derived/calculated columns and apply condition-based filtering to a dataset.
- Exposure to business-intelligence dashboarding and calculated (DAX) measures using Power BI.
- Experience carrying a real transactional dataset through a complete analytics workflow — cleaning, validation, exploratory analysis, and reporting.
- Improved ability to translate analytical findings into clear, actionable, stakeholder- oriented recommendations.

4.10 Certificates of Completion and Other Work Proof

The Internship Completion Certificate issued by BeeSkilled, confirming successful completion of the 6-week Data Analyst internship, is presented in the preliminary pages of this report (page iii). Figure 4.9 below reproduces the original internship offer letter issued at the start of the program as supporting communication proof.



Figure 4.9: BeeSkilled Internship Offer Letter

No additional communication or completion emails beyond the documents shown above were available at the time of preparing this report.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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BeeSkilled. Internship Offer Letter issued to Adamya Dixit, 17 July 2026.

BeeSkilled. Internship Completion Certificate issued to Adamya Dixit, 28 August 2026.

BeeSkilled Data Analyst Internship curriculum materials — Week 1 to Week 4 program content, provided by BeeSkilled.